

JOLIE TRAN

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PROFESSIONAL SUMMARY

Data Scientist specializing in data science, geospatial analysis, and software-driven workflow automation. Proven experience designing production-grade data pipelines, analytical frameworks, and user-facing applications that enable planners, engineers, and decision-makers to efficiently leverage mobility, transit, traffic, and safety datasets. Combines expertise in Python, R, SQL, GIS, and data visualization with a strong transportation planning background to deliver scalable solutions that improve operational efficiency, accelerate project delivery, and support data-driven investments.

TECHNICAL SKILLS

- Data & Scripting: R, Python, SQL (PostgreSQL, PostGIS), PL/SQL
- Web Development: HTML, CSS, JavaScript, LeafletJS, MapboxGL, ChartJS, JSON, Shiny, Streamlit

EXPERIENCE

Data Scientist	WSP USA Inc.	April 2024 – Present
• Built and deployed Python- and R-based automation tools with Shiny/Streamlit front-ends and standalone executables to convert raw Synchro, IDAX, and VISSIM outputs into analysis-ready datasets, cutting processing time for engineering/planning teams from ~2 days to under 15 minutes per project. • Designed ETL pipelines (Python) integrating GTFS, Replica, INRIX, Swiftly, crash, and spatial datasets into standardized multimodal databases, replacing manual spreadsheet workflows and cutting data-prep time by ~80% across 15+ studies. • Replaced a legacy Excel-based reporting workflow for WSDOT tolling operations with a custom Streamlit dashboard (Python/Pandas backend), automating data ingestion and report generation and cutting manual analyst effort by ~80%. • Built a fully automated crash analysis workflow: from data cleaning through statistical analysis to visualization, using WSDOT crash data, designed to scale across jurisdictions, turning multi-week manual analyses into a process completed in hours. • Led crash and safety data analysis for 4+ regional safety action plans, building geospatial models and interactive mapping tools, including customized solutions for Tribal agencies, to identify high-crash corridors and prioritize safety investments.		
Data Scientist, Part-time	Washington State Transportation Center (TRAC)	Jun 2023 – Jan 2025
• Developed PostgreSQL/PostGIS-based ETL workflows for large-scale transit and sidewalk datasets, delivering 90%+ precision in automated spatial integration and validation. • Implemented GTFS-driven correction logic to resolve mis-encoded ORCA transit stops, ensuring analytical consistency across ridership and service metrics. • Built multi-layered dashboards and analytical pipelines correlating ORCA ridership, service reliability, and disruption events to identify key behavioral and operational drivers. • Engineered SQL procedures for spatial data quality control, including topology validation, missing network generation, and cross-source conflict resolution in OSM. • Designed a Leaflet-based web application for public data review and feedback, improving transparency and iterative data quality improvements in equity-focused sidewalk mapping. • Validated and benchmarked solutions across multi-city deployments, enabling scalable adoption beyond Seattle.		
Social Media Researcher, Intern	Humanities Data Science Summer Institute @ UW	Jun 2023 – Aug 2023
• Collaborated with a multidisciplinary team to conduct a comprehensive analysis of social media conversations on Twitter, focusing on the influence of historical figures in political movements and historical events. • Analyzed an extensive dataset of 3 million tweets spanning from 2007 to 2023, applying advanced data processing techniques using Pandas and Dask libraries in Python. This efficient analysis revealed key trends and patterns. • Incorporated a mixed-method research approach, combining quantitative data analysis with qualitative assessment to uncover nuanced patterns and trends in social media conversations. • Contributed valuable insights and visualized compelling data for an ongoing book project led by a professor at UW, enhancing the research and its potential impact.		

EDUCATION

M.S. in Human-Centered Design and Engineering

University of Washington, Seattle

Sep 2026 – Jun 2028

B.A. in Geography: Data Science

University of Washington, Seattle

Sep 2021 – Mar 2024

- In-major GPA: 4.0. Cumulative GPA: 3.94
- Coursework: Data Structures and Algorithm, Database Management, Machine Learning, Web Development

VOLUNTEER EXPERIENCE

Undergraduate Research Assistant

Laboratory for Auditory Neuroscience and Development

Apr 2023 – Mar 2024

- Utilized MATLAB and the Hilbert transform to extract the envelope of the speech from EEG data of 7-month-old and 11-month-old infants, with a success rate of 70%.
- Engaged in the pre-processing and artifact removal pipeline, achieving a 25% reduction in data distortion.
- Analyzed chance prediction accuracy and determined whether the observed prediction accuracy exceeded chance